

FoldShield⁺⁺ Structural Analysis Report

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Input Structures

Reference PDB	1AKE.pdb
Query PDB	4AKE.pdb
Mode	ca
Ref chain	A
Query chain	A

Classical Coordinate Metrics

These metrics saturate near 1.0 even when function diverges. Use FoldShield++ signals below for structural perturbation insight.

Metric	Value	Note
TM-score	0.700	>=0.50 same fold · >=0.90 nearly identical
RMSD (A)	3.600	Lower is better · >3.0 A = significant deviation

FoldShield++ Signal Scores — Phase 1 + Phase 2

All active signals with scores, fusion weights, and interpretation. Phase 2 signals are highlighted in blue.

Braid similarity	0.805	0.20	strong braid similarity
Motif similarity (UL-DSL)	0.712	0.37	moderate motif similarity
SSEF entropy (Phase 2) [P2]	0.911	0.20	symbolic grammar nearly identical — buried-core mutation or identical
Persistent homology (Phase 3) [P2]	0.845	0.23	similar topological persistence structure

Fusion Layer v2 — Combined Score

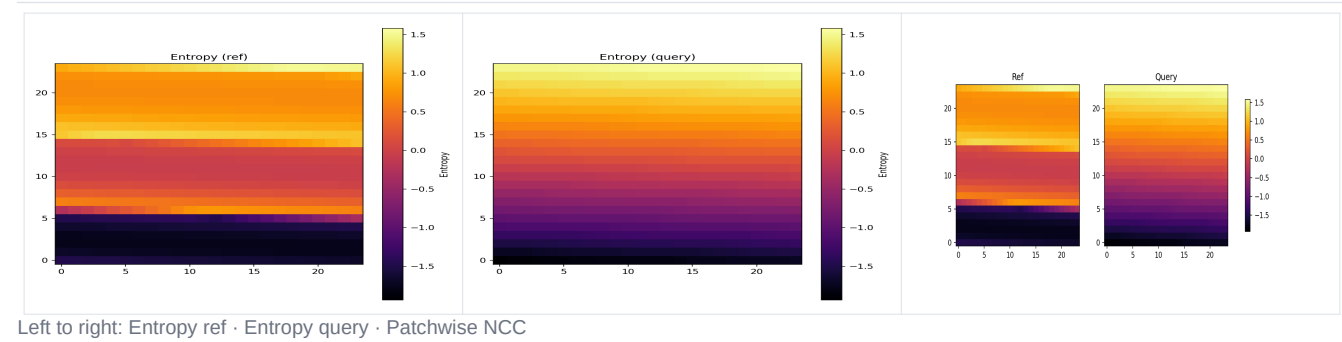
0.8014	phase2	highly similar structural architecture

Phase 2 — Perturbation Classification

SSEF score	Local topology	Gap score	Perturbation class
0.911	0.466	0.446	Conformational transition

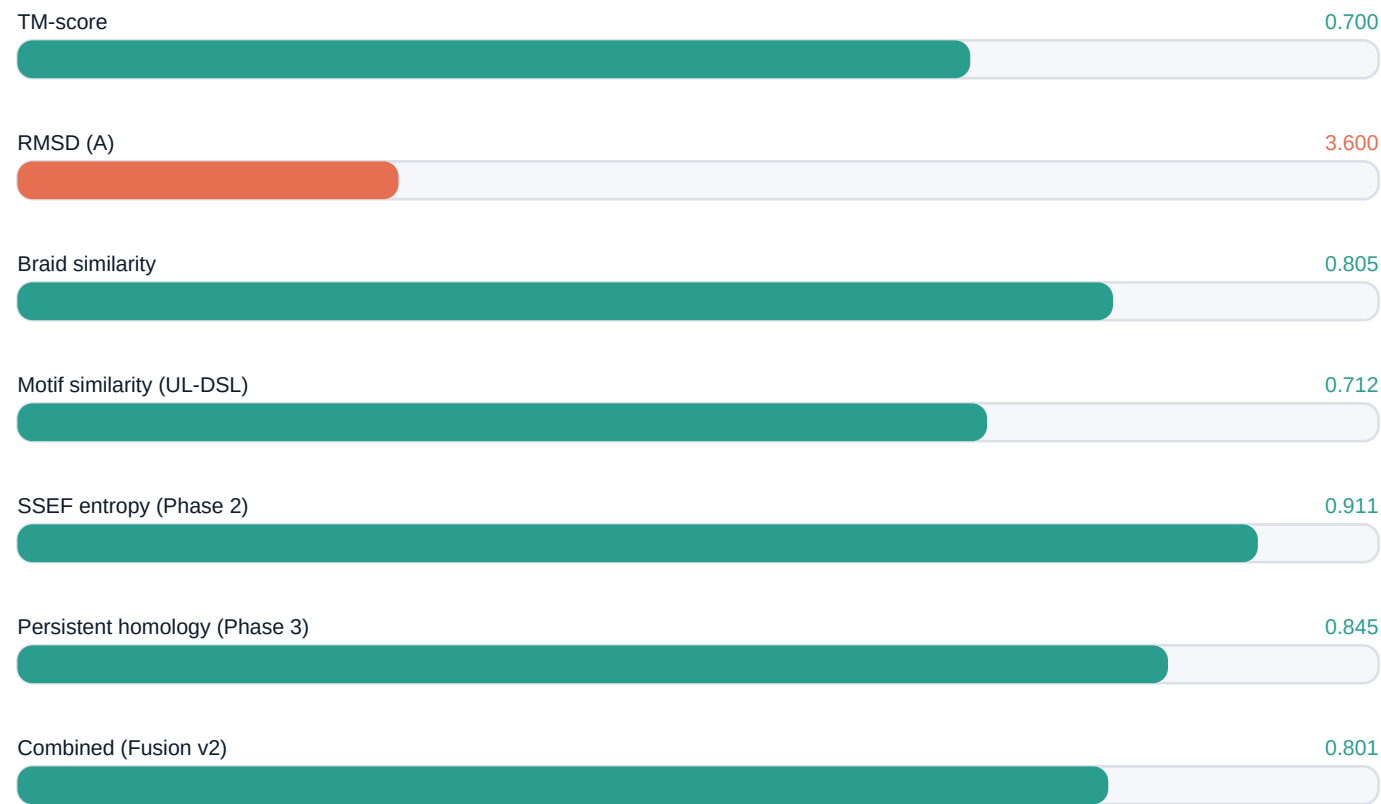
Gap = SSEF minus local topology. <0.15 buried-core mutation | 0.15-0.36 fold-switch | >0.36 conformational transition

Entropy Maps & Signal Profiles



Signal Score Bars

Visual 0-1 scale for all active signals. Green ≥ 0.66 / Amber 0.33-0.66 / Red < 0.33 .



Symbolic Score Detail

Symbolic Similarity Score	0.805	overall nearly identical
Behavioural Correlation Similarity	0.771	entropy maps strongly correlated
Structural Topology Similarity	1.000	identical set of generators
Structural Topology Ordering Similarity	0.638	moderate pattern similarity

Signal Analysis — Figures

